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Project Title:	STUDY OF FLUID FLOW CONTROL IN PROTEIN CRYSTALLIZATION USING STRONG MAGNETIC FIELDS		
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Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
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Zip Code:	35806	**Congressional District:**	5
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	An important component in biotechnology, particularly in the area of protein engineering and rational drug design is the knowledge of the precise three-dimensional molecular structure of proteins. The quality of structural information obtained from X-ray diffraction methods is directly dependent on the degree of perfection of the protein crystals. As a consequence, the growth of high quality macromolecular crystals for diffraction analyses has been the central focus for biochemists, biologists, and bioengineers. To date, space protein crystallization experiments have yielded inconclusive results with ~20% of flight-grown proteins showing promising results. The space crystal growth environment offers reduced fluid motion – convection due solutal buoyancy and sedimentation effects. We postulate that we can reduce convection in ground-based experiments and we can understand the different aspects of convection control through the use of strong magnetic fields and field gradients. In 2004, progress was made on several fronts in the project. A key finding based on our measurements of the density and susceptibility variation of protein solutions as a function of protein concentration showed that an equivalent g-level of only ~0.2 go (where go is terrestrial gravity) is realizable in a practical, high field strength superconducting magnet. This is because the protein solution is weakly diamagnetic (repelled by a magnetic field) and requires a large magnetic field gradient to affect the solutal buoyancy driven flow. We developed a novel flow control methodology wherein we make the protein solution paramagnetic (attracted to a magnetic field) by adding paramagnetic salts. Fairly high paramagnetism can be induced by adding small quantities of the salt. We successfully grew protein crystals from these solutions and developed the theory for magnetic flow control. Validation experiments are planned for early 2005. A major advantage of the technique is that we can now do the magnetic flow control experiments in-house at NASA Marshall Space Flight Center using the available 5 Tesla magnet. The research on the project has resulted in several presentations, papers, and awards. We also developed key collaborations in applying our research to artificial gravity applications for supporting NASA's extended duration exploration missions. The concept is to use magnetic levitation as a microgravity model for cell cultures in order to determine the gravity threshold levels and artificial gravity requirements for exploration purposes.
Task Description:	Task Objectives Specific goals for the investigation are: 1. To develop an understanding of convection control in diamagnetic fluids with concentration gradients through careful experimentation. Specifically solutal buoyancy driven convection due to crystal growth will be considered. Strong magnetic fields and field gradients will be utilized to unambiguously resolve the role of fluid flow in the process. 2. To develop predictive measures for successful crystallization in a magnetic field using analyses and numerical modeling for use in future protein crystal growth experiments. This will establish criteria that can be used to estimate the efficacy of magnetic field flow damping on crystallization of candidate proteins. 3. To demonstrate the understanding of convection damping by high magnetic fields to a class of proteins that is of interest and whose structure is as yet not determined. 4. To compare quantitatively, the quality of the grown crystals with and without a magnetic field. X-ray diffraction techniques will be used for the comparative studies.
	Prior ambiguities of space-based protein crystallization need clarification. Understanding the fundamentals of a process is critical to major innovations and breakthroughs. Protein crystallization is however a complex process involving the interactions of several physical phenomena such as kinetics, fluid flow, physical chemistry, self-assembly, etc. and is still an empirical science. It holds the promise of the development of next generation designer drugs for and hence it is vigorously pursued by the drug industry. Protein crystallization experiments have flown on more space missions than any other targeted research effort but a comprehensive understanding of the fundamental processes and the beneficial effects of the microgravity environment are not clear. This investigation uses magnetic fields to counter gravitational effects – a terrestrial simulation of microgravity, in order to investigate crystal growth phenomena in reduced gravity. The current working hypothesis is that convection is detrimental to macromolecular crystal growth, although the exact mechanisms are unclear, and yet only 20% of space-grown proteins have shown results better than ground-based studies. This investigation will clarify the impact of fluid flow during the growth process either as a causative factor of some undesirable effects or point to the need to investigate other (non-flow) effects to improve crystal quality. The study adopts a stepping stone approach to the problem by validating theory by simple experiments before moving on to full-fledged protein crystallization. It represents an important contribution in the understanding of the complex crystallization phenomena.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	To date, on this project, a novel, proven technique of obtaining zero-gravity conditions, terrestrially, for protein crystallization conditions has been successfully demonstrated. Protein crystals grown in reduced gravity realize the so-called diffusion limited (no convection) growth conditions. Both solutal buoyancy driven convection (flows due to density differences based on concentration) and sedimentation are reduced significantly, thereby providing a quiescent environment for crystal growth. There is evidence from past missions that some proteins grown in space (~20%) are of better quality than terrestrially grown crystals. Crystallization experiments in space flight are expensive, opportunities are sparse, and are not amenable to a quick-turnaround iterative process that terrestrial experimentation can provide. The new technology developed in this investigation can provide protein crystal growth conditions similar to that realized during space experimentation. The technique is based on using strong magnetic fields to achieve the desired results.
	In this project we sought to critically examine the fluid flow effects due to solutal convection and develop a clear understanding of flow control through the use of magnetic fields and field gradients. Using analyses, numerical computations, and experimental work, we developed fluid equilibrium and stability conditions for the use of magnetic forces in protein crystal growth studies. Our NASA supported work, for the first time, clearly elucidates the scientific principles underlying the technique of magnetic controlled concentration gradient driven convection and provides the detailed procedure for realizing diffusion-limited growth conditions in a laboratory. A direct result from the studies is the development of a technique to simulate variable gravity conditions (zero to reduced gravity to hypergravity) terrestrially using superconducting magnets. This opens up a very enabling technique for protein crystallization experiments that is easily accessible, tunable, affording iterative capability, and above all more economical than space flight. The method can also be readily extended to other phenomena, such as, cell culture experiments to investigate variable gravity and artificial gravity effects. Our work has been published in peer reviewed journals and also presented in national and international conferences. A new technology disclosure was accorded to the work as well as the NASA Marshall Space Flight Center Director's Commendation.

Task Progress:	A) Typically, protein crystal growth solutions are diamagnetic in nature. While tailoring the diamagnetic forces in a magnetic field is the frequently utilized mechanism for fluid flow control in protein crystallization, the required magnetic force for the effort is not attainable in easily accessible magnets. The difference in magnetic susceptibilities of the protein and growth solution is small and hence considerable magnetic force is required to attenuate or halt solutal convection. Our measurements showed that only a few facilities exist in the world that can be utilized for the purpose. Additionally, since protein crystal growth occurs slowly, over several days to weeks, maintaining the high magnetic field continuously presents a formidable challenge. B) Our work with weakly paramagnetic fluids surrounded by a diamagnetic fluid, showed that a magnetic field can be used to halt vertical or horizontal fluid motion due to solutal convection locally. The experimental work confirmed our theoretical predictions that a fluid parcel surrounded by another fluid of less susceptibility is not stable at the equilibrium points implying that complete cessation of flows in such a system using a magnetic field and field gradients is not possible. C) Protein crystal growth involves the use of a precipitating agent. Various salts and compounds have been used for this purpose. Using a paramagnetic salt in the crystallization process makes the magnetic convection control technique amenable to readily available lower strength superconducting magnets. Changing the magnetic properties of the growth media requires a totally different experiment configuration and stability studies. Our work has shown that that solutal convection in protein crystallization can be stabilized (complete fluid flow cessation) for the duration of the growth process if the surrounding fluid (growth media) has a larger magnetic susceptibility than the crystallizing protein and the magnetic field has a specific structure. This has been an important finding and innovation of this investigation. Our experiments and theoretical predictions are in good agreement. This finding has applications potential beyond protein crystallization studies. D) While other investigators using the same concept have reported conflicting results (both beneficial and non-beneficial), we find that these studies did not carefully evaluate and understand all of the underlying physics associated with using magnetic fields and magnetically responsive growth media. Our systematic approach to the problem has led to important discoveries and contributions to the understanding of fluid convection control during protein crystallization. We have developed detailed procedures for protein crystallization experiments using magnetic fields. A direct result from the studies is the development of a technique to simulate variable gravity conditions (zero to reduced gravity to hypergravity) terrestrially using superconducting magnets for studying different phenomena. E) We have developed protein crystal growth procedures using a paramagnetic salt as the precipitating agent. Our work has shown the feasibility of using paramagnetic salts of differing magnetic susceptibilities in the crystal growth process and developed optimum crystallization protocols for a benchmark protein. This implies that candidate proteins, whose structures must be determined, can be first tested for crystallization conditions using different salts with higher magnetic susceptibility and then our technique can be utilized to realize diffusion limited growth for crystal growth and subsequent characterization.
Bibliography Type:	Description: (Last Updated: 03/17/2017)
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